

Math 104 – Real Analysis

Homework 5 — my solutions

Fall 2026 ■ Keshav Ramamurthy

Homework 5

Deadline: October 4

Write $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$; subsequential limits may be infinite.

Problem 1.

Define (s_n) by

$$s_{3k-2} = -2 + \frac{1}{k}, \quad s_{3k-1} = 1 - \frac{1}{k}, \quad s_{3k} = k \quad (k \in \mathbb{N}).$$

Find all subsequential limits in $\overline{\mathbb{R}}$, and determine $\liminf s_n$ and $\limsup s_n$. State whether (s_n) has a limit in $\overline{\mathbb{R}}$.

Problem 2.

Let (a_n) and (b_n) be bounded sequences, and let $r_n \geq 0$ with $r_n \rightarrow 0$. Suppose $a_n \leq b_n + r_n$ for all sufficiently large n . Prove that

$$\liminf a_n \leq \liminf b_n, \quad \limsup a_n \leq \limsup b_n.$$

Work directly with infima and suprema of sufficiently late tails; do not assume that (a_n) or (b_n) has an ordinary limit.

Problem 3.

Let (p_n) and (q_n) repeat the following blocks of four:

$$(p_n) = (1, -1, 3, 0, 1, -1, 3, 0, \dots), \\ (q_n) = (0, 4, -2, 2, 0, 4, -2, 2, \dots).$$

Set $u_n = p_n + 1/n$ and $v_n = q_n - 2/n$. Compute

$$(a) \liminf u_n + \liminf v_n, \quad (b) \liminf(u_n + v_n), \quad (c) \liminf u_n + \limsup v_n, \\ (d) \limsup(u_n + v_n), \quad (e) \limsup u_n + \limsup v_n.$$

There is no need to write a proof.

Problem 4.

Determine whether each series converges or diverges. In each case, name a valid criterion and supply the inequality or limit that makes it apply. There is no need to find an exact sum.

$$(a) \sum_{n=1}^{\infty} \frac{n+1}{n^2+3n+1}, \quad (b) \sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1},$$

Problem 5.

Let (b_n) be a real sequence with $0 < c \leq |b_n| \leq C$ for all sufficiently large n , where c, C are fixed constants. Prove that $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if $\sum_{n=1}^{\infty} a_n b_n$ does, for every real sequence (a_n) .

Show that the positive lower bound is essential: give bounded (b_n) with $b_n \neq 0$ for every n and a real sequence (a_n) such that one of these series converges absolutely and the other does not. Justify both claims.

Solutions

Problem 1

Solution. The terms split into three subsequences:

$$s_{3k-2} \longrightarrow -2, \quad s_{3k-1} \longrightarrow 1, \quad s_{3k} \longrightarrow +\infty.$$

These are all the subsequential limits. Therefore,

$$\liminf_{n \rightarrow \infty} s_n = -2, \quad \limsup_{n \rightarrow \infty} s_n = +\infty.$$

The subsequences tending to -2 and 1 have different limits, so (s_n) does not have a limit in $\mathbb{R}$. $\square$

Problem 2

Solution. For N large enough, $a_n \leq b_n + r_n$ holds for every $n \geq N$. Set

$$R_N = \sup_{n \geq N} r_n.$$

Since $r_n \rightarrow 0$ and $r_n \geq 0$, we have $R_N \rightarrow 0$. For every $n \geq N$, $a_n \leq b_n + R_N$. Taking the infimum and supremum over this same tail gives

$$\inf_{n \geq N} a_n \leq \inf_{n \geq N} b_n + R_N, \quad \sup_{n \geq N} a_n \leq \sup_{n \geq N} b_n + R_N.$$

As $N \rightarrow \infty$, the tail infima and suprema approach the corresponding lim infs and lim sups, while $R_N \rightarrow 0$. Thus

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} b_n, \quad \limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} b_n.$$

$\square$

Problem 3

Solution. The terms $1/n$ and $2/n$ tend to zero, so the repeating blocks determine the lim infs and lim sups. For $u_n + v_n$, the block is $(1, 3, 1, 2)$. Therefore,

- (a) $\liminf u_n + \liminf v_n = -1 + (-2) = -3$,
- (b) $\liminf(u_n + v_n) = 1$,
- (c) $\liminf u_n + \limsup v_n = -1 + 4 = 3$,
- (d) $\limsup(u_n + v_n) = 3$,
- (e) $\limsup u_n + \limsup v_n = 3 + 4 = 7$.

$\square$

Problem 4

Solution. For (a), the terms are positive and

$$\frac{n+1}{n^2+3n+1} > \frac{1}{n+3},$$

since $(n+1)(n+3) > n^2+3n+1$. The series $\sum_{n=1}^{\infty} 1/(n+3)$ is a shifted harmonic series and diverges, so the comparison test shows that the given series diverges.

For (b), the terms do not tend to zero:

$$\left| (-1)^n \frac{n}{n+1} \right| = \frac{n}{n+1} \rightarrow 1.$$

By the nth-term test, this series diverges. □

Problem 5

Solution. We will first show the direction that $\sum_{i=1}^n a_i$ converges implies $\sum_{i=1}^n a_i b_i$ converges given that

$$0 < c < |b_n| \leq C$$

for all sufficiently large n .

Then, notice that (for sufficiently large n)

$$\sum_{i=1}^n a_i \leq \sum_{i=1}^n a_i C \leq C \sum_{i=1}^n a_i$$

and similarly:

$$c \sum_{i=1}^n a_i \leq \sum_{i=1}^n a_i b_i \leq C \sum_{i=1}^n a_i$$

Both $c \sum a_i$ and $C \sum a_i$ converge, then $\sum a_i b_i$ is bounded and must converge. To show the converse, we start from $\sum a_i b_i$ converging, and we wish to prove that this implies $\sum a_i$ must converge. In order to do so, we again bound it like the previous step, noticing for sufficiently large n that

$$\sum a_i c \leq \sum a_i b_i \leq a_i C$$

where $c, C > 0$ □